

July 14, 2026

Journal of Machine Learning Research

Dear Editors:

We are writing to submit our manuscript “When to Trust Kernel Change? A Predictive Validation Framework for Representation Diagnostics” to the *Journal of Machine Learning Research*.

Our manuscript studies whether representation similarity metrics (CKA, SVCCA, PWCCA, Frobenius distance) are reliable indicators of generalization in neural networks. We propose the RDEP framework for validating representation diagnostics, demonstrate that kernel change magnitude alone is neither necessary nor sufficient for predicting generalization improvement, and identify three reliability regimes across controlled synthetic and real-world experiments. The manuscript includes a finite-sample theoretical analysis of the Diagnostic Validation Gap (Proposition 1) bounding the variance of cross-validated diagnostic correlations.

Previous publications. This manuscript has not been previously published, nor is it an extended version of any earlier conference paper. No version of this work has appeared in any peer-reviewed venue.

Conflict of interest declaration. The author declares no conflicts of interest. As a single-author submission, there are no co-authors with potential competing interests. None of the suggested action editors (Fukumizu, Rosasco, Steinwart, Sriperumbudur, d’Alche-Buc) have recently collaborated with the author. To the best of my knowledge, none of the suggested action editors or reviewers have a conflict of interest with the author. In particular, there is no advisee/advisor, co-author, or institutional relationship between the author and any suggested individual.

Author consent. As the sole author, I confirm that I am fully aware of this submission and consent to its review by the *Journal of Machine Learning Research*.

Suggested Action Editors:

- Kenji Fukumizu, The Institute of Statistical Mathematics, Japan
- Lorenzo Rosasco, University of Genova, Italy and MIT, USA
- Ingo Steinwart, University of Stuttgart, Germany
- Bharath Sriperumbudur, Pennsylvania State University, USA
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Suggested Reviewers:

- Carlo Ciliberto, University College London, UK (kernel methods, statistical learning theory, structured prediction)
- Alessandro Rudi, INRIA, France (kernel methods, statistical machine learning)

- Taiji Suzuki, University of Tokyo, Japan (kernel methods, deep learning, optimization)
- Tengyuan Liang, University of Chicago, USA (deep learning theory, kernel methods, interpolation, high-dimensional asymptotics)
- Behnam Neyshabur, Google Research, USA (deep learning, learning theory, generalization)

Keywords: representation similarity, diagnostic validation, feature learning, representation evaluation, neural tangent kernel

Sincerely,

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